1 Visual region extraction

Patch–box intersection-area weights
224 × 224 · no CLS token



Input image I

CLIP ViT-B/16 visual encoder ❄

Patch features
14 × 14 × 768

YOLO detector ❄

Boxes B
(x_1, y_1, x_2, y_2)

Box-weighted ROI pooling

Region descriptors
N × 768

Linear projection 🔥
768 → 512

V

2 Semantic bank–region alignment

P_n · 150 × 512
A

V_1, T_1

Object fusion 1 🔥
B

Object fusion 2 🔥
B

V_o

V : N × 512

P_v · 150 × 512
A

Relation fusion 🔥
B

V_r

A TRAIN-only bank preparation & aggregation

B One shared fusion detail; separate parameters

3 Image-specific prompt–region alignment

UPT boxes/classes + YOLO boxes/classes

Class-consistent matching · IoU ≥ 0.5
ROI index i / ordered pair (s, o)

Unmatched slots are omitted.

t_o / t_r + ROI links feed the corresponding Slot MLP.

Object adapter · C 🔥
t_o + linked ROI metadata

Relation adapter · C 🔥
t_r + linked endpoint metadata

V_o'

V_r'

I → UPT detector ❄
same input image

Object / relation
phrases

CLIP text encoder ❄
t_o / t_r

Named ports → C

4 Caption generation

Parallel decoder · D 🔥
V_o' and V_r' remain separate
3 layers · dimension 512 · heads 8
Prefix → next-token distribution

Word embedding + fixed PE

5 Caption reranking

Beam search · width 5
max 20 new tokens

Five candidate captions

Three-score reranking · E
S = 0.3D + 0.4C + 0.3O

Final caption

A TRAIN banks & semantic aggregation

TRAIN captions
train + restval

spaCy POS
+ lemmatization

Noun / verb
lemmas

CLIP text encoder ❄

OFFLINE · BUILT ONCE

TRAIN references only

No VALID / TEST references

Shared banks ≠ image-specific prompts

150 soft prototypes per bank;
not 150 selected words.

Fixed noun bank E_n
512-D embeddings

Fixed verb bank E_v
512-D embeddings

Learned semantic 🔥
aggregation

Learned semantic 🔥
aggregation

P_n
150 × 512

P_v
150 × 512

Learned aggregation · separate noun / verb parameters

E

Linear W_a 🔥

MLP 🔥

Softmax
over bank entries

Weights A

Linear W_b 🔥

Transformed embeddings
E W_b

A^T × E W_b

P : 150 × 512

A = softmax_bank(MLP(EW_a)) P = A^T(EW_b)

B Semantic fusion block

Same structure, separate parameters · two outputs: X′ and T

Regions X

Prototypes P

Cosine similarity
X and P

Masked region softmax
valid regions · τ = 0.1

Prototype relevance
$\Sigma_i \alpha_{ki}$ cosine$_{ki}$

Softmax over prototypes
Relevance weights β

Soft weighting; no hard top-K.

T = β ⊙ P
Soft semantic features

X

Multi-head self-attn 🔥
Q = K = V = X

Add & Norm 🔥

Feed Forward 🔥

Add & Norm 🔥

H

Multi-head cross-attn 🔥
Q = H · K = V = T

Add & Norm 🔥

Feed Forward 🔥

Add & Norm 🔥

C

[H; C]

Learned sigmoid gate 🔥
g = σ(Wg[H; C])

Weighted fusion
g ⊙ H + (1 − g) ⊙ C

Updated regions X′

Object fusion 1: (X, P) = (V, P_n) → (V_1, T_1)
Object fusion 2: (V_1, T_1) → V_o Relation fusion: (V_o, P_v) → V_r

C Image-specific prompt extraction & ROI matching

Image I
same input

UPT interaction detector ❄

Object / relation phrases
a bicycle / a person riding a bicycle

CLIP text encoder ❄

UPT boxes / classes

Class-consistent matching
IoU ≥ 0.5

Object ROI i
Ordered relation pair (s, o)

YOLO boxes / classes

Unmatched slots are omitted.

Object t_o / Relation t_r
512-D text embeddings

C Two image-specific prompt adapters

Both slot builders use original V_o. [;] denotes concatenation.

Object adapter

$[t_o; V_o[i]; b_i]$
Original region feature · normalized box

Concatenation

Slot MLP 🔥
Linear → GELU → Linear

Object slots S_o

V_o

Masked multi-head cross-attn 🔥
Q = V_o · K = V = S_o
Mask: linked ROI i only

Output projection W_o 🔥
Zero-initialized

+

V_o' : N × 512

Relation adapter

$[t_r; V_o[s]; V_o[o]; b_s; b_o]$
Ordered endpoints · normalized boxes

Concatenation

Slot MLP 🔥
Linear → GELU → Linear

Relation slots S_r

V_r

Masked multi-head cross-attn 🔥
Q = V_r · K = V = S_r
Mask: endpoints s or o only

Output projection W_r 🔥
Zero-initialized

+

V_r' : N × 512

No valid linked slot → zero residual. No adapter gate.

Visual detail Box-weighted ROI pooling

Patch features f_j
14 × 14 × 768 · no CLS

Bounding box B_i
normalized to patch grid

Patch–box intersection area
w_{ij} = area(B_i ∩ patch$_j$)

Weighted patch mean
$r_i = \Sigma_j w_{ij} f_j / \Sigma_j w_{ij}$

Region descriptors
N × 768

Linear projection 🔥
768 → 512

V
N × 512

One weighted descriptor per box; partial patch overlaps contribute proportionally. No crop image encoder.

D Caption generation

Token prefix

Word embedding 🔥
raw e

√512 e

Fixed sinusoidal position
PE

One decoder layer ×3 · dimension 512 · heads 8

Causal masked self-attn 🔥
Q = K = V = layer input

Add & Norm 🔥

H

V_o'

Object cross-attn 🔥
Q = H · K = V = V_o'

Relation cross-attn 🔥
Q = H · K = V = V_r'

V_r'

Gates use raw word embedding e, without PE.

Learned sigmoid gate 🔥
g_o = σ(W_o[C_o; e])

Learned sigmoid gate 🔥
g_r = σ(W_r[C_r; e])

Gated sum z
g_o ⊙ C_o + g_r ⊙ C_r

Feed Forward 🔥

Add & Norm 🔥

No extra H residual into the gated sum.

Vocabulary projection 🔥

Softmax

Autoregressive beam search
width 5 · max 20 new tokens

Five candidate captions
y_1 … y_5

Generated token → next prefix

E Caption reranking

Same five candidates y_1 … y_5

D · Decoder likelihood

Sum token log-probabilities
Raw sequence log-probability

Min–max normalization
over the same 5 candidates

No length normalization.

D

C · Image–text alignment

CLIP image / text encoders ❄
Image I + candidate caption

Cosine-based CLIPScore
2.5 max(cosine, 0)

Min–max normalization
over the same 5 candidates

C

O · Object consistency

Mentioned object extraction
from candidate caption

Supported object labels
from YOLO detections

Detector-based
object consistency

Normalized object score
(consistency + 1) / 2

O

S(y) = 0.3 D(y) + 0.4 C(y) + 0.3 O(y)

Select highest-scoring candidate

Final caption

Equal scores: earlier beam wins.

References and CHAIR are not used for selection.

F Training only

Stage 1 · Cross-entropy training

TRAIN reference caption

Shift right

Current decoder 🔥

Predicted token distribution

Target tokens

LCE

Stage 2 · SCST + cross-entropy

Current model · same image 🔥

Sampled caption y_s

Greedy caption yg

TRAIN references

CIDEr reward r(·)

Current-batch LCE
TRAIN teacher forcing

Advantage: r(y_s) − r(yg)

Rollout log p(y_s)

LSCST · detached advantage

Total loss = LCE + 0.05 LSCST

References: training and evaluation only.

❄ Frozen parameters 🔥 Trainable parameters

Rounded: processing · Square: data · [;]: concat · +: residual sum

Orange: object semantics · Blue: relation semantics · Gray-blue: visual / decoder